

Olist E-Commerce SQL Data Analysis

Business Insight & Recommendation

I. Data Overview

The dataset comes from Olist, a Brazilian e-commerce platform. It contains multiple tables including *customers*, *geolocation*, *orders*, *payments*, *products*, *orders*, *sellers*, *order items* and *order reviews*. The data spans of two years (2017-2018), allowing analysis of customer behavior and revenue monthly trends. Overall, the dataset is clean and relational, though some tables require joins to consolidate insights.

II. Three Business Insights

1. **High-value customers are few but critical (Q1 & Q6):** The top 10 customers by spending contribute disproportionately to revenue. Segmentation into spend tiers shows that the “High” tier is small but generates the majority of sales. This means customer value is highly concentrated.
2. **Revenue is seasonal and fluctuates (Q2 & Q3):** Monthly revenue trends reveal peaks and troughs, with noticeable month-over-month changes. This indicates seasonality in shopping behavior, suggesting that promotions and campaigns should align with high-revenue months.
3. **Product categories drive concentration risk (Q4, Q5, Q8):** A handful of categories dominate revenue, and the top category alone contributes a significant percentage of total revenue. Within categories, the top 3 products account for most sales. This shows strong dependence on a narrow set of products and categories.

III. Recommendations

1. **Strengthen retention strategies:** Since most customers are one-time buyers (Q7), Olist should invest in loyalty programs, personalized offers, and post-purchase engagement to convert them into repeat buyers. This would stabilize revenue and reduce reliance on new customer acquisition.
2. **Diversify product focus:** Given that revenue is concentrated in a few categories (Q4 & Q8), Olist should expand marketing and inventory support for mid-tier categories. This reduces risk if demand shifts away from the current top category and creates new growth opportunities.